Narendra Sir's Typed Notes

Animal Kingdom — PU-II Biology (Chapter 4)

Exam-ready study notes • Read with the chapter sequence • Use the recap terms for rapid revision.

1. Basis of Classification

Rapid recap: Levels of organisation • Symmetry • Germ layers • Coelom • Segmentation • Notochord

Why Classification Matters

Over a million species of animals have been described so far, making classification essential. Basis of classification includes: levels of organisation, patterns of digestive/circulatory/reproductive systems, body symmetry, arrangement of cells in germ layers, nature of coelom, segmentation, and presence or absence of notochord.

Criteria Used in Animal Classification

levels of organisation: cellular level (e.g. Porifera), tissue level (e.g. Coelenterata), organ level (e.g. Platyhelminthes), and organ system level (e.g. Echinoderms and Chordates). Organ system pattern may be incomplete (one opening, e.g. Platyhelminthes) or complete (separate mouth and anus, with open or closed circulation).

2. Body Symmetry, Germ Layers and Coelom

Rapid recap: Radial • Bilateral • Asymmetry • Diploblastic • Triploblastic • Coelomate/Pseudocoelomate/Acoelomate

Body Symmetry

radial symmetry: any plane through the central axis divides the body into identical halves (e.g. coelenterates, ctenophores, echinoderms). Bilateral symmetry: only one plane divides the body into identical left and right halves (e.g. insects, chordates).

Germ Layers

Asymmetry: no plane divides the body into equal halves (e.g. sponges/Porifera). Diploblastic animals have ectoderm and endoderm with mesoglea in between (e.g. coelenterates); triploblastic animals have ectoderm, mesoderm and endoderm (e.g. Platyhelminthes to Chordates). Coelomates possess a mesoderm-lined body cavity (coelom) — e.g. Annelida, Mollusca, Arthropoda, Echinodermata, Hemichordata, Chordata.

Body Cavity

Pseudocoelomates have mesoderm as scattered pouches between ectoderm and endoderm — e.g. Aschelminthes. Acoelomates lack a body cavity altogether — e.g. Platyhelminthes.

Quick comparison: Body Symmetry, Germ Layers and Coelom

Feature	Example phyla
Cellular level of organisation	Porifera
Tissue level of organisation	Coelenterata, Ctenophora
Organ level of organisation	Platyhelminthes
Organ system level of organisation	Aschelminthes to Chordata

3. Segmentation and Notochord

Rapid recap: Metamerism • Chordates • Non-chordates

Metameric Segmentation

Segmentation: body externally and internally divided into segments with repeated organs, called metameric segmentation or metamerism (e.g. Earthworm). Notochord is a mesoderm-derived rod-like structure formed dorsally during embryonic development.

The Notochord

Animals with notochord are chordates; animals without notochord are non-chordates (e.g. Porifera to Echinodermata).

4. Phylum Porifera

Rapid recap: Sponges • Cellular level • Canal system • Ostia/Spongocoel/Osculum

Organisation and Water Canal System

Commonly called sponges; generally marine, primitive, multicellular, asymmetrical, with cellular level of organisation. Have a water transport / canal system: water enters through ostia into the spongocoel and leaves through the osculum, helping in food gathering, gas exchange and excretion.

Skeleton and Reproduction

Spongocoel and canals are lined by collar cells (choanocytes); digestion is intracellular. Skeleton consists of spicules or spongin fibres.

Examples

Reproduction: asexual (fragmentation, budding, gemmules) or sexual (gametes); sponges are hermaphrodite/monoecious with internal fertilisation and indirect development (amphiblastula larva in Sycon, parenchymula larva in Leucosolenia). Examples: Sycon, Spongilla (fresh water sponge), Euspongia (Bath sponge).

5. Phylum Coelenterata (Cnidaria)

Rapid recap: Cnidoblasts • Polyp and medusa • Metagenesis

Organisation and Cnidoblasts

Coelenterates/Cnidaria are aquatic (mostly marine), sessile or free-swimming, radially symmetrical, diploblastic, with tissue level of organisation. Have a central gastro-vascular cavity (GVC) with a single opening, the hypostome, surrounded by tentacles.

Forms, Digestion and Reproduction

Tentacles have cnidoblasts/cnidocytes containing nematocysts, used for anchorage, defence and capturing prey — the name Cnidaria comes from cnidoblasts. Digestion can be extracellular and intracellular; skeleton (in corals) is composed of calcium carbonate. Two basic body forms: polyp (sessile, cylindrical, e.g. Hydra, Adamsia) and medusa (umbrella-shaped, free-swimming, e.g. Aurelia/jellyfish).

Examples

Cnidarians showing both forms exhibit alternation of generation (metagenesis): polyps produce medusae asexually, medusae form polyps sexually (e.g. Obelia). Examples: Physalia (Portuguese man of war), Adamsia (sea anemone), Pennatula (sea-pen), Gorgonia (sea-fan), Meandrina (brain coral), Obelia (sea fur), Aurelia (jellyfish).

6. Phylum Ctenophora

Rapid recap: Comb jellies • Bioluminescence • Comb plates

Key Organisation Features

Commonly known as sea walnuts, comb jellies or sea gooseberries; exclusively marine, radially symmetrical, diploblastic, tissue level of organisation. Digestion is extracellular or intracellular; locomotion is by eight rows of externally located ciliated comb plates.

Comb Plates, Colloblasts and Reproduction

Sexes are not separate (hermaphrodite); reproduction is sexual only with external fertilisation and indirect development (cydippid larva).

Examples

Exhibit bioluminescence — the ability to emit light. Examples: Pleurobrachia (sea gooseberry) and Ctenoplana (sea walnut).

7. Phylum Platyhelminthes

Rapid recap: Flatworms • Flame cells • Hermaphrodite

Body Plan and Organisation

Commonly called flatworms due to dorso-ventrally flattened bodies; bilaterally symmetrical, triploblastic, acoelomate, organ level of organisation. Nutrition is parasitic (endoparasites) with hooks and suckers present, or nutrients absorbed directly from the host.

Parasitic Adaptations and Examples

Excretion through specialised flame cells/solenocytes/protonephridia, which aid osmoregulation and excretion. Reproduction sexual and asexual (fission); Platyhelminthes are hermaphrodites with internal fertilisation and development consisting of several larval stages; some show high regeneration capacity. Examples: Taenia (tapeworm, parasite in pigs and humans), Fasciola (liver fluke, parasite in goat/sheep), Dugesia (Planaria, aquatic free-living).

8. Phylum Aschelminthes

Rapid recap: Round worms • Pseudocoelomates • Dioecious

Body Plan and Digestion

Known as round worms due to circular cross-section body; bilaterally symmetrical, triploblastic, pseudocoelomate, organ system level of organisation. May be free-living (aquatic/terrestrial) or parasitic on plants and animals; alimentary canal is complete with a well-developed muscular pharynx.

Reproduction and Examples

Excretory tube removes waste through the excretory pore. Aschelminthes are dioecious (sexes separate), with females often longer than males; fertilisation is internal, development direct or indirect. Examples: Ascaris (round worm, causes Ascariasis), Wuchereria (filarial worm, causes Filariasis/Elephantiasis transmitted by mosquitoes — Anopheles in Africa, Culex in America), Ancylostoma (hookworm, infects small intestine of cats and dogs).

9. Phylum Annelida

Rapid recap: Segmented worms • Nephridia • Closed circulation

Body Organisation and Segmentation

May be aquatic (marine/freshwater) or terrestrial; bilaterally symmetrical, triploblastic, coelomate, organ system level, with metamerically segmented bodies. Locomotion by longitudinal and circular muscles; parapodia present in aquatic forms for swimming.

Locomotion, Excretion and Examples

Circulatory system closed; excretory organs are nephridia; nervous system has paired ganglia connected to a double ventral nerve cord by lateral nerves. Reproduction is strictly sexual; some forms dioecious (e.g. Nereis) while some are monoecious/hermaphrodite (e.g. Earthworm and Leech). Examples: Nereis (clam worm), Pheretima (earthworm), Hirudinaria (leech).

10. Phylum Arthropoda

Rapid recap: Largest phylum • Jointed appendages • Open circulation • Malpighian tubules

Body Plan and Organisation

largest phylum in Animalia, comprising over two-thirds of known animal species, mostly insects; bilaterally symmetrical, triploblastic, coelomate (haemocoel — coelom filled with blood). Body covered by chitinous exoskeleton with distinct head, thorax and abdomen; have jointed appendages. Respiration through gills/book gills (aquatic) or book lungs/tracheal system (terrestrial); circulatory system is open; excretory organs are Malpighian tubules.

Respiration, Excretion and Examples

Sensory organs include compound and simple eyes, antennae and statocysts (balance organs). Mostly dioecious with internal fertilisation and direct or indirect development. Examples: Apis (honey bee), Bombyx (silkworm), Laccifer (lac insect); vectors like Aedes, Anopheles, Culex; pests like Locusta and cockroach; Limulus (king crab) is a living fossil.

11. Phylum Mollusca

Rapid recap: Second largest phylum • Mantle cavity • Radula

Typical Molluscan Body Plan

second largest phylum in Animalia among invertebrates; terrestrial or aquatic; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body covered by a calcareous shell with distinct head, muscular foot and visceral hump; a soft mantle covers the visceral hump. Mantle cavity (space between hump and mantle) contains feather-like gills with respiratory and excretory functions.

Physiology and Examples

Excretory organs: one or two pairs of kidneys called metanephridia; sensory tentacles on the head; feeding organ is the file-like radula. Molluscs are dioecious and oviparous, with direct or indirect development. Examples: Pila (apple snail), Pinctada (pearl oyster), Octopus (devil fish), Sepia (cuttlefish), Loligo (squid), Aplysia (sea-hare), Dentalium (tusk shell), Chaetopleura (chiton).

12. Phylum Echinodermata

Rapid recap: Spiny-bodied • Water vascular system • Marine

Body Plan and Water Vascular System

Echinodermata means spiny-bodied due to an endoskeleton of calcareous ossicles; exclusively marine; bilaterally symmetrical, triploblastic, coelomate, organ system level. Digestive system is complete with mouth on the ventral side and anus on the dorsal side.

Reproduction and Examples

Water vascular / ambulacral system is a distinctive feature, helping in locomotion, food capture/transport and respiration. Reproduction is sexual with external fertilisation and indirect development (Bipinnaria larva). Examples: Asterias (star fish), Echinus (sea urchin), Cucumaria (sea cucumber), Ophiura (brittle star), Antedon (sea lily).

13. Phylum Hemichordata

Rapid recap: Worm-like marine animals • Proboscis, collar, trunk

Body Plan and Affinities

Earlier placed as a sub-phylum of Chordata; a small group of worm-like marine animals; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body is cylindrical, composed of anterior proboscis, collar and a long trunk.

Reproduction and Examples

Circulatory system open; respiratory organs are gills; excretory organ is the proboscis gland. Sexes separate, fertilisation external, development indirect. Examples: Balanoglossus (tongue worm/acorn worm) and Saccoglossus.

14. Phylum Chordata — General Features and Subphyla

Rapid recap: Notochord • Dorsal nerve cord • Gill slits • Post-anal tail

Defining Chordate Features

Fundamental characters of Chordata: presence of notochord, dorsal hollow nerve cord, and paired pharyngeal gill slits. Bilaterally symmetrical, triploblastic, coelomate, organ system level; possess a post-anal tail and a closed circulatory system.

Chordates and Non-chordates

Phylum Chordata is divided into three subphyla: Urochordata/Tunicata (e.g. Ascidia, Salpa, Doliolum, Herdmania), Cephalochordata (e.g. Branchiostoma/Amphioxus or Lancelet), and Vertebrata/Craniata.

The Three Subphyla

"All vertebrates are chordates, but all chordates are not vertebrates" — in Vertebrata, the embryonic notochord is replaced by a cartilaginous or bony vertebral column in adults. Vertebrata is divided into Agnatha (lacks jaw — Class Cyclostomata) and Gnathostomata (bears jaw), which is further split into superclass Pisces (bear fins — Chondrichthyes, Osteichthyes) and superclass Tetrapoda (bear limbs — Amphibia, Reptilia, Aves, Mammalia).

Quick comparison: Phylum Chordata

Chordates	Non-chordates
Notochord present	Notochord absent
Central nervous system dorsal, hollow and single	Central nervous system ventral, solid and double
Pharynx perforated by gill slits	Gill slits absent
Heart is ventral	Heart is dorsal (if present)
A post-anal part (tail) is present	Post-anal tail is absent

15. Class Cyclostomata

Rapid recap: Jawless • Ectoparasitic • Cartilaginous

General Features

Sucking circular mouth without jaws; elongated, eel-like body. Marine but migrate to fresh water for spawning; die shortly after spawning; larvae metamorphose and return to the ocean.

Life History and Examples

Ectoparasitic on other fishes; respiration by 6-15 pairs of gill slits; body devoid of scales and paired fins; cranium and vertebral column are cartilaginous. Examples: Petromyzon (lamprey), Myxine (hag fish).

16. Class Chondrichthyes

Rapid recap: Cartilaginous fish • Placoid scales • Poikilothermous

Cartilaginous Fish Features

Marine habitat; cartilaginous endoskeleton, streamlined body; notochord persists throughout life. Skin has small placoid scales; predaceous with strong jaws, mouth located ventrally, teeth are modified placoid scales directed backward. Respiration by 5-7 pairs of gills without operculum; must swim continuously to avoid sinking (no air bladder); heart two-chambered (1 auricle, 1 ventricle).

Physiology and Examples

Some have electric organs (e.g. Torpedo, 8-220 volts) or poisonous stings (e.g. Trygon/sting ray); poikilothermous. Sexes separate; males have claspers on pelvic fins; internal fertilisation; many are viviparous. Examples: Scoliodon (dog fish), Pristis (saw fish), Carcharodon (great white shark), Trygon (sting ray).

17. Class Osteichthyes

Rapid recap: Bony fish • Operculum • Air bladder

Bony Fish Features

Marine and freshwater habitats; bony endoskeleton, streamlined body with terminal mouth. Body covered by cycloid or ctenoid scales; respiration by four pairs of gills covered by operculum; air bladder provides buoyancy.

Reproduction and Examples

Heart two-chambered; poikilothermous; sexes separate, external fertilisation, mostly oviparous with direct development. Examples: Marine — Exocoetus (flying fish), Hippocampus (sea horse); Freshwater — Labeo (rohu), Catla, Clarias (magur); Aquarium — Betta (fighting fish), Pterophyllum (angel fish).

Quick comparison: Class Osteichthyes

Chondrichthyes	Osteichthyes
Skeleton made of cartilage	Skeleton made of bones
Exclusively marine	Marine and freshwater forms
Possess placoid scales	Possess cycloid or ctenoid scales
Mouth on ventral side	Mouth terminal in position
5 pairs of gill slits, no operculum	4 pairs of gill slits with operculum
Heterocercal caudal tail	Homocercal caudal tail
Devoid of air bladder	Possess air bladder
Males possess claspers	Claspers absent in males

18. Class Amphibia

Rapid recap: Water and land • Cloaca • Three-chambered heart

Adaptations to Two Habitats

Can live both in water and on land; body divisible into head and trunk with two pairs of limbs; tail may be present in some. Alimentary canal, excretory and reproductive tracts open into a common cloaca; skin is moist without scales. Eyes have eyelids; ear represented by tympanum; respiration by gills, lungs and skin.

Reproduction and Examples

Heart three-chambered (two auricles, one ventricle); poikilothermous. Sexes separate, external fertilisation, oviparous, indirect development. Examples: Bufo (toad), Rana (frog), Salamandra (salamander), Hyla (tree frog), Ichthyopis (limbless amphibian).

19. Class Reptilia

Rapid recap: Creeping/crawling • Dry cornified skin • Oviparous

Terrestrial Adaptations

Characterised by creeping/crawling locomotion; mostly terrestrial; body covered by dry cornified skin with epidermal scales/scutes. No external ears but tympanum present; two pairs of limbs when present; heart three-chambered (four-chambered in crocodiles); poikilothermous.

Reproduction and Examples

Snakes and lizards shed their scales as skin cast. Sexes separate; internal fertilisation; oviparous; direct development. Examples: Chelone (turtle), Testudo (tortoise), Chameleon, Calotes (garden lizard), Crocodilus, Alligator, Hemidactylus (wall lizard); poisonous snakes — Naja (cobra), Bangarus (krait), Vipera (viper).

20. Class Aves

Rapid recap: Wings and flight • Four-chambered heart • Homoiothermous

Flight Adaptations

Characterised by wings and flight ability (except flightless birds like ostrich); have beaks; forelimbs modified into wings, hind limbs used for walking/swimming/clasping branches. Skin is dry with oil glands only at the base of the tail; endoskeleton ossified with hollow, pneumatic long bones.

Reproduction and Examples

Respiration by lungs and air sacs; digestive system has crop and gizzard as additional chambers; heart four-chambered. Homoiothermous (warm-blooded); internal fertilisation, oviparous, direct development, sexes separate. Examples: Corvus (crow), Columba (pigeon), Psittacula (parrot), Struthio (ostrich), Pavo (peacock), Aptenodytes (penguin), Neophron (vulture).

21. Class Mammalia

Rapid recap: Mammary glands • Hair • Viviparous

Diagnostic Mammalian Features

Found in a wide variety of habitats — polar ice, deserts, mountains, forests, grasslands, dark caves; adapted to fly, swim or walk. Presence of mammary glands (producing milk for young) is a unique mammalian character; respiration by lungs.

Reproduction and Examples

Two pairs of limbs adapted for walking, running, climbing, burrowing, swimming or flying; skin covered by hair; external ears (pinnae) present. Heart four-chambered; homoiothermous; sexes separate; internal fertilisation; mostly viviparous (exception: Platypus is oviparous); direct development. Examples: Ornithorhynchus (platypus, oviparous), Macropus (kangaroo), Pteropus (flying fox), Camelus (camel), Macaca (monkey), Rattus (rat), Canis (dog), Felis (cat), Elephas (elephant), Equus (horse), Delphinus (dolphin), Balaenoptera (blue whale), Panthera tigris (tiger), Panthera leo (lion).

22. Summary — Salient Features of Phyla

Rapid recap: Quick comparison table

Comparison at a Glance

This summary table brings together level of organisation, symmetry, coelom, segmentation, digestive/circulatory/respiratory systems, and distinctive features for all eleven phyla covered in this chapter — ideal for quick pre-exam revision.

Exam Focus

Use the salient comparisons in this topic to distinguish groups accurately in classification questions.

Quick comparison: Summary — Salient Features of Phyla

Phylum	Organisation	Symmetry	Coelom	Distinctive Feature
Porifera	Cellular	Many/None	Absent	Body with pores and canals in walls
Coelenterata (Cnidaria)	Tissue	Radial	Absent	Cnidoblasts present
Ctenophora	Tissue	Radial	Absent	Comb plates for locomotion
Platyhelminthes	Organ	Bilateral	Absent (acoelomate)	Flat body, suckers
Aschelminthes	Organ system	Bilateral	Pseudocoelomate	Often worm-shaped, elongated
Annelida	Organ system	Bilateral	Coelomate	Body segmented like rings
Arthropoda	Organ system	Bilateral	Coelomate	Exoskeleton of cuticle, jointed appendages
Mollusca	Organ system	Bilateral	Coelomate	External shell usually present

Phylum	Organisation	Symmetry	Coelom	Distinctive Feature
Echinodermata	Organ system	Radial	Coelomate	Water vascular system, radial symmetry
Hemichordata	Organ system	Bilateral	Coelomate	Worm-like with proboscis, collar and trunk
Chordata	Organ system	Bilateral	Coelomate	Notochord, dorsal hollow nerve cord, gill slits with limbs or fins